

AIVOA.AI

Database Schema & Data Model

PostgreSQL / Supabase design for the connected QMS lifecycle

Document 3 of 5

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AIVOA.AI — ROUND 2

File 2 — Database Schema & Data Model

Detailed PostgreSQL / Supabase design for upgrading the existing Round 1 PharmaOne-AI system into a connected QMS lifecycle

Challenge alignment: The supplied brief requires an API manufacturing example covering Raw Material → Manufacturing → In-Process Checks → Deviation → Investigation → CAPA → Batch Release → Complaint/Recall → Supplier Quality, and specifically emphasizes Investigation → Root Cause → CAPA → Closure. The schema below is intentionally centered on that chain rather than a full enterprise QMS.

1. Existing Round 1 Database: What You Already Have

The current project already has an Alembic/PostgreSQL foundation. The important existing tables are listed below. Round 2 should extend these tables rather than replacing them.

1.1 Important current design detail

The existing Deviation model already stores the final user-reviewed impact/severity separately from AI recommendations and AI extraction/assessment snapshots. That is the correct pattern to preserve in Round 2: authoritative QMS data belongs to the reviewed record, while AI output is stored as traceable evidence/recommendation history.

Existing Deviation concept deviations ■■■■ final reviewed fields ■ ■■■■ impact ■ ■■■■ severity ■ ■■■■ assessment_reason ■ ■■■■ status ■ ■■■■ AI snapshots ■■■■ ai_extraction ■■■■ ai_assessment ■■■■ ai_recommended_impact ■■■■ ai_recommended_severity ■■■■ ai_reason ■■■■ ai_evidence

2. Target Round 2 Data Model

Company ■■■■ Site ■■■■ Batch ■ ■■■■ Batch Material ■ ■ ■■■■ Raw Material ■■■■ Supplier ■ ■■■■ Manufacturing Step ■ ■■■■ In-Process Check ■ ■ ■■■■ Deviation ■ ■■■■ Batch Release ■ ■■■■ Deviation ■■■■ Investigation ■ ■■■■ Investigation Task ■ ■■■■ Evidence ■ ■■■■ Root Cause Analysis ■ ■■■■ Conclusion ■ ■■■■ CAPA ■■■■ CAPA Action ■■■■ Effectiveness Check Batch ■■■■ Complaint ■■■■ Recall / Adverse Event links Deviation ■■■■ Supplier Quality Event (when supplier-related) Any quality entity ■■■■ Audit Event Any AI-assisted action ■■■■ AI Interaction ■■■■ AI Evidence / Source References

The database is deliberately relational for traceability. JSONB is used only where the shape is genuinely variable, such as AI snapshots, extracted payloads, flexible metadata and evidence details. Core business relationships should not be hidden inside JSON.

3. Naming and Data Conventions

4. Manufacturing Context Tables

4.1 batches

4.2 raw_materials

4.3 batch_materials

Join table connecting a batch to the raw materials actually used. This is what lets a supplier be traced into a manufacturing batch and then into a deviation.

4.4 manufacturing_steps

4.5 in_process_checks

5. Deviation Data Model — Round 1 Extension

Do not create a second deviation table. Extend the existing deviations table with explicit foreign keys to the manufacturing context and lifecycle ownership.

5.1 Keep existing fields

reference, source, reported_by, occurred_on, detected_on

company_id, site_id, department, responsible_team

product_name, product_code, batch_number

manufacturing_stage, equipment, title, description, deviation_type

expected_condition, actual_condition, duration, parameter

immediate_action, batch_status, qa_notified

impact, severity, assessment_reason

AI recommendation/evidence snapshots

6. Investigation Schema

6.1 investigations

6.2 investigation_tasks

6.3 investigation_evidence

7. Root Cause Analysis Schema

7.1 root_cause_analyses

Example why_chain JSONB [{"step": 1, "question": "Why did the deviation occur?", "answer": "Cooling response was delayed."}, {"step": 2, "question": "Why was cooling response delayed?", "answer": "Cooling valve did not respond correctly."}, {"step": 3, "question": "Why did the valve fail?", "answer": "Valve actuator malfunctioned."}, {"step": 4, "question": "Why was the actuator condition not known?", "answer": "Maintenance failed to detect degradation."}]

8. CAPA Schema

8.1 capas

8.2 capa_actions

8.3 effectiveness_checks

9. Batch Release / Disposition

10. Complaints, Recalls and Adverse Events

For the challenge, these are context modules. Keep their schemas intentionally small and focus on traceability back to the batch/deviation.

10.1 complaints

10.2 recalls

10.3 adverse_events

11. Supplier Quality Schema

11.1 suppliers

11.2 supplier_quality_events

12. AI and RAG Persistence

12.1 ai_interactions

12.2 ai_evidence_links

The existing knowledge_documents and knowledge_chunks tables remain the RAG corpus. ai_evidence_links makes the runtime relationship explicit so a reviewer can see exactly which source chunks supported a recommendation.

13. Audit Trail Extension

The existing audit_events table should remain the central append-only event stream. Extend the AuditAction vocabulary rather than creating separate audit tables for every module.

Suggested audit actions DEVIATION_CREATED DEVIATION_UPDATED DEVIATION_PROCESSED DEVIATION_STATUS_CHANGED DEVIATION_ASSIGNED INVESTIGATION_CREATED INVESTIGATION_TASK_CREATED INVESTIGATION_TASK_COMPLETED INVESTIGATION_EVIDENCE_ADDED ROOT_CAUSE_DRAFTED ROOT_CAUSE_ACCEPTED ROOT_CAUSE_EDITED CAPA_CREATED CAPA_ACTION_CREATED CAPA_ACTION_COMPLETED CAPA_EFFECTIVENESS_RECORDED CAPA_CLOSED CAPA_REOPENED BATCH_RELEASE_REVIEWED BATCH_RELEASE_APPROVED BATCH_RELEASE_REJECTED COMPLAINT_CREATED RECALL_CREATED SUPPLIER_EVENT_CREATED AI_RECOMMENDATION_ACCEPTED AI_RECOMMENDATION_EDITED AI_RECOMMENDATION_REJECTED

13.1 Audit meta example

```
{ "request_id": "uuid", "old_status": "under_investigation", "new_status": "capa_required", "reason": "Investigation completed; CAPA required.", "source": "web", "ai_interaction_id": "uuid", "changed_fields": ["status"] }
```

14. Relationship and Foreign-Key Rules

For a regulated-quality-style demo, the safest mental model is: quality records are not ordinary disposable CRUD rows. Use status transitions, archival/deactivation and audit history instead of hard deletion.

15. Index Strategy

16. Database Constraints and Integrity Rules

Batch number should be unique within a company/site as appropriate for the demo.

Investigation reference, CAPA reference, complaint reference and recall reference must be unique.

An investigation must reference an existing deviation.

A CAPA created from the workflow must reference the source deviation and investigation.

A root-cause analysis must belong to one investigation.

A CAPA effectiveness check belongs to one CAPA.

A batch release decision must reference a batch and the authorized reviewer.

Foreign-key relationships should be used for core traceability instead of copying only text labels.

Status values should be validated at the application boundary and, where practical, with CHECK constraints.

Dates should be logically checked where practical: due_date should not precede created_at for normal records; completed_at should not precede created_at.

Closed records should not be directly edited through normal update paths. The database does not replace service-layer workflow checks, but constraints should prevent obviously invalid relationships.

17. Alembic Migration Plan

Your current project already uses Alembic. Round 2 should continue that migration chain. Do not manually edit an already-applied migration in a shared environment.

0005_round2_manufacturing_context ■■■■ companies/sites already exist ■■■■ create suppliers ■■■■ create raw_materials ■■■■ create batches ■■■■ create batch_materials ■■■■ create manufacturing_steps ■■■■ create in_process_checks 0006_round2_deviation_links ■■■■ add batch_id to deviations ■■■■ add manufacturing_step_id ■■■■ add in_process_check_id ■■■■ add owner_id ■■■■ add closure fields ■■■■ add indexes/FKs 0007_round2_investigation ■■■■ investigations ■■■■ investigation_tasks ■■■■ investigation_evidence ■■■■ root_cause_analyses 0008_round2_capa ■■■■ capas ■■■■ capa_actions ■■■■ effectiveness_checks 0009_round2_release_complaint_supplier ■■■■ batch_release_reviews ■■■■ complaints ■■■■ recalls ■■■■ adverse_events ■■■■ supplier_quality_events 0010_round2_ai_audit ■■■■ ai_interactions ■■■■ ai_evidence_links ■■■■ expanded audit action vocabulary

18. Demo Seed Data for the Round 2 Story

Seed one connected story instead of many unrelated records. This makes the UI demo fast and makes every module explainable.

19. Queries the UI Will Need

```
-- Deviation detail page SELECT d.*, b.batch_number, ipc.parameter, ipc.actual_value FROM deviations d LEFT JOIN batches b ON b.id = d.batch_id LEFT JOIN in_process_checks ipc ON ipc.id = d.in_process_check_id WHERE d.id = :deviation_id AND d.company_id = :company_id; -- Investigation + root cause + CAPA SELECT i.*, r.final_root_cause, c.reference AS capa_reference, c.status AS capa_status FROM investigations i LEFT JOIN root_cause_analyses r ON r.investigation_id = i.id LEFT JOIN capas c ON c.investigation_id = i.id WHERE i.deviation_id = :deviation_id AND i.company_id = :company_id; -- Dashboard open counts SELECT status, count(*) FROM deviations WHERE company_id = :company_id GROUP BY status; SELECT status, count(*) FROM capas WHERE company_id = :company_id GROUP BY status;
```

19.1 Important query rule

Never fetch a record by ID alone when the user is authenticated in a tenant-scoped application. The query should include the server-derived company_id/site authorization context, or use a repository method that guarantees tenant filtering.

20. Supabase/PostgreSQL RLS Direction

The current application already performs authorization/tenant checks in the backend. If Supabase RLS is enabled for direct client access, mirror the same company membership rule at the database layer. Do not rely on frontend filtering.

Conceptual policy: User → company_memberships → company ↓ row.company_id A user can SELECT/UPDATE a row only when an active membership exists for row.company_id. For site-scoped roles: row.site_id must also match the membership site_id, unless the role is explicitly company-wide.

For the fast demo, keep direct browser access to sensitive business tables limited and route quality operations through FastAPI. This avoids duplicating complex business workflow rules in the client.

21. Performance and Practicality

Keep dashboard queries aggregate-oriented and indexed.

Do not load full evidence documents into the dashboard. Load counts/metadata and fetch content on demand.

Do not store large PDF/image binaries in PostgreSQL. Store files in controlled object storage and keep references/metadata in the DB.

Store only short evidence snippets in ai_evidence_links; keep the canonical source in the knowledge/document store.

Paginate deviations, investigations, CAPAs, complaints and audit events.

Use eager loading only for the small set of linked records needed by a detail page.

22. What to Actually Implement Under Time Pressure

22.1 Minimal real database path

companies ↓ batches ↓ in_process_checks ↓ deviations ↓ investigations ↓ root_cause_analyses ↓ capas ↓ capa_actions ↓ effectiveness_checks ↓ batch_release_reviews

23. Final Data-Model Rule

If an evaluator asks “How do you know this deviation led to that CAPA?”, the database should answer without relying on text matching. The chain should be explicit foreign-key relationships: deviation_id → investigation.deviation_id → root_cause_analyses.investigation_id → capas.investigation_id/deviation_id → capa_actions.capa_id → effectiveness_checks.capa_id. Batch, process check, supplier and complaint records should be connected in the same way.

END OF FILE 2

Layer	Purpose	Round 2 decision
Existing foundation	companies, sites, profiles, memberships, deviations, audit events, knowledge documents/chunks	Preserve
Manufacturing context	batches, raw materials, suppliers, process steps and in-process checks	Add lightweight
Core quality lifecycle	investigations, root-cause analysis, CAPA, actions, effectiveness, closure	Build deeply
Downstream context	batch release, complaints, recalls/adverse-event links, supplier quality events	Build light
AI governance	recommendations, evidence, human decision history, model/prompt metadata	Add for traceability

Current table	Purpose	Important columns
companies	Tenant/organization	id, name, code, status, created_at, updated_at
sites	Manufacturing/lab site	id, company_id, name, code, status
profiles	User profile linked to Supabase Auth	id, email, full_name, employee_id, department, job_title, status
company_memberships	User-to-company/site role	id, user_id, company_id, site_id, role, is_active
deviations	Round 1 core quality event	id, reference, source, company_id, site_id, product, batch, manufacturing_stage, equipment, title, description, expected/actual condition, immediate_action, batch_status, impact, severity, status, AI snapshots/evidence
audit_events	Existing audit trail	id, action, entity_type, entity_id, user_id, company_id, meta, timestamps

Current table	Purpose	Important columns
knowledge_documents	RAG source document metadata	id, title, filename, source, doc_type, checksum, status, chunk_count, meta
knowledge_chunks	RAG chunks and embeddings	id, document_id, chunk_index, chunk_id, content, embedding, meta

Convention	Decision	Reason
Internal PK	UUID	Safe for distributed creation and consistent with existing code
Human reference	DEV-2026-018, INV-2026-012, CAPA-2026-009	Readable identifiers for the UI/demo
Tenant key	company_id	Every business table should carry or inherit a company boundary
Site key	site_id where applicable	Supports multiple manufacturing/lab sites
Time	TIMESTAMPTZ	Auditability and consistent ordering
Enums	VARCHAR + application enum or PostgreSQL enum only if stable	Existing project uses non-native SQLAlchemy enums for portability
Flexible evidence	JSONB	AI/RAG and evidence payloads vary
Soft deletion	Avoid for GMP records	Use status/archival and audit trail instead of deleting quality records
History	Append audit events	Do not overwrite important decisions without history
Indexes	Tenant + status + reference + foreign keys	Optimizes dashboard and record navigation

Column	Type	Null	Description
id	UUID PK	NO	Internal batch ID
company_id	UUID FK → companies.id	NO	Tenant boundary
site_id	UUID FK → sites.id	NO	Manufacturing site
batch_number	VARCHAR(120)	NO	Human batch/lot number
product_name	VARCHAR(200)	NO	API/product name
product_code	VARCHAR(120)	YES	Internal product code
status	VARCHAR(32)	NO	planned/in_progress/completed/on_hold/quarantined/released/rejected
manufacturing_start	TIMESTAMPTZ	YES	Start
manufacturing_end	TIMESTAMPTZ	YES	End
recipe_version	VARCHAR(64)	YES	Approved process/recipe version
release_status	VARCHAR(32)	NO	pending/released/quarantined/rejected
meta	JSONB	YES	Demo metadata
created_at	TIMESTAMPTZ	NO	Timestamp
updated_at	TIMESTAMPTZ	NO	Timestamp

Column	Type	Null	Description
id	UUID PK	NO	Material ID
company_id	UUID FK	NO	Tenant
material_code	VARCHAR(120)	NO	Material code
material_name	VARCHAR(200)	NO	Material name
supplier_id	UUID FK	YES	Approved supplier
status	VARCHAR(32)	NO	approved/quarantine/rejected/under_test
specification	TEXT	YES	Quality specification
coa_reference	VARCHAR(255)	YES	Certificate of analysis reference
meta	JSONB	YES	Additional material metadata
created_at	TIMESTAMPTZ	NO	Timestamp
updated_at	TIMESTAMPTZ	NO	Timestamp

Column	Type	Null	Description
id	UUID PK	NO	Join ID
batch_id	UUID FK → batches.id	NO	Manufacturing batch

Column	Type	Null	Description
material_id	UUID FK → raw_materials.id	NO	Raw material
material_lot_number	VARCHAR(120)	YES	Actual material lot
quantity	NUMERIC(18,6)	YES	Quantity used
unit	VARCHAR(32)	YES	Unit
qc_status	VARCHAR(32)	NO	pending/pass/fail
used_at	TIMESTAMPTZ	YES	Usage timestamp

Column	Type	Null	Description
id	UUID PK	NO	Step ID
batch_id	UUID FK	NO	Parent batch
step_number	INTEGER	NO	Sequence
name	VARCHAR(160)	NO	e.g. Reaction
status	VARCHAR(32)	NO	not_started/in_progress/completed/blocked
equipment	VARCHAR(160)	YES	Equipment used
started_at	TIMESTAMPTZ	YES	Start
completed_at	TIMESTAMPTZ	YES	End

Column	Type	Null	Description
id	UUID PK	NO	Check ID
batch_id	UUID FK	NO	Parent batch
manufacturing_step_id	UUID FK	YES	Step where check occurred
parameter	VARCHAR(160)	NO	Temperature, pressure, pH, etc.
specification	TEXT	YES	Expected range/criteria
actual_value	VARCHAR(160)	YES	Observed value
unit	VARCHAR(32)	YES	Unit
status	VARCHAR(32)	NO	in_spec/out_of_spec/warning
checked_at	TIMESTAMPTZ	NO	Measurement time
source_reference	VARCHAR(255)	YES	Instrument/log reference
meta	JSONB	YES	Instrument/test metadata

New/changed column	Type	Purpose
batch_id	UUID FK → batches.id	Strong relationship instead of only storing batch_number text
manufacturing_step_id	UUID FK	Exact process step
in_process_check_id	UUID FK	Exact measurement that triggered the deviation
owner_id	UUID FK → profiles.id	Responsible quality/investigation owner
workflow_version	VARCHAR(32)	Version of lifecycle rules used
closed_at	TIMESTAMPTZ	Closure timestamp
closed_by	UUID FK → profiles.id	Authorized closer
closure_reason	TEXT	Final closure rationale

Column	Type	Null	Description
id	UUID PK	NO	Investigation ID
reference	VARCHAR(32) UNIQUE	NO	INV-2026-012
company_id	UUID FK	NO	Tenant
site_id	UUID FK	YES	Site
deviation_id	UUID FK → deviations.id	NO	Source deviation
owner_id	UUID FK → profiles.id	NO	Investigator/owner
status	VARCHAR(32)	NO	draft/in_progress/root_cause_pending/completed
scope	TEXT	YES	Investigation scope
conclusion	TEXT	YES	Final conclusion

Column	Type	Null	Description
product_impact_assessment	TEXT	YES	Impact assessment
completed_at	TIMESTAMPTZ	YES	Completion time
completed_by	UUID FK	YES	Completing user
created_at	TIMESTAMPTZ	NO	Timestamp
updated_at	TIMESTAMPTZ	NO	Timestamp

Column	Type	Description
id	UUID PK	Task ID
investigation_id	UUID FK	Parent investigation
title	VARCHAR(255)	Task
description	TEXT	Task details
owner_id	UUID FK	Responsible person
due_date	DATE	Due date
status	VARCHAR(32)	not_started/in_progress/completed/blocked
completed_at	TIMESTAMPTZ	Completion
evidence_count	INTEGER	UI convenience/count

Column	Type	Description
id	UUID PK	Evidence ID
investigation_id	UUID FK	Parent
evidence_type	VARCHAR(48)	batch_record/equipment_log/interview/lab_result/document/photo/other
title	VARCHAR(255)	Display name
storage_object_id	VARCHAR(512)	Controlled storage reference
knowledge_document_id	UUID FK nullable	Link to RAG source when applicable
notes	TEXT	Investigator notes
uploaded_by	UUID FK	Actor
created_at	TIMESTAMPTZ	Timestamp

Column	Type	Description
id	UUID PK	Root-cause record
investigation_id	UUID FK UNIQUE	One primary RCA per investigation for demo
method	VARCHAR(32)	5_whys/fishbone/other
problem_statement	TEXT	Starting problem
why_chain	JSONB	Ordered Why 1..N entries
final_root_cause	TEXT	Confirmed root cause
contributing_factors	JSONB	Array/list of contributing factors
evidence_summary	TEXT	Evidence supporting root cause
ai_draft	JSONB	AI suggestion snapshot
human_decision	VARCHAR(32)	accepted/edited/rejected
reviewed_by	UUID FK	Human reviewer
reviewed_at	TIMESTAMPTZ	Review timestamp
created_at	TIMESTAMPTZ	Timestamp
updated_at	TIMESTAMPTZ	Timestamp

Column	Type	Description
id	UUID PK	CAPA ID
reference	VARCHAR(32) UNIQUE	CAPA-2026-009
company_id	UUID FK	Tenant
site_id	UUID FK	Site
deviation_id	UUID FK	Source deviation
investigation_id	UUID FK	Source investigation

Column	Type	Description
root_cause_analysis_id	UUID FK	Confirmed root cause
owner_id	UUID FK	CAPA owner
status	VARCHAR(32)	draft/in_progress/effectiveness_pending/ready_for_closure/closed/ineffective
problem_statement	TEXT	Problem being addressed
root_cause	TEXT	Root cause being addressed
priority	VARCHAR(24)	Priority
due_date	DATE	Overall CAPA due date
closed_at	TIMESTAMPTZ	Closure
closed_by	UUID FK	Closer
closure_reason	TEXT	Closure rationale

Column	Type	Description
id	UUID PK	Action ID
capa_id	UUID FK	Parent CAPA
action_type	VARCHAR(24)	corrective/preventive
title	VARCHAR(255)	Action
description	TEXT	Action detail
owner_id	UUID FK	Owner
due_date	DATE	Due
status	VARCHAR(32)	not_started/in_progress/completed/overdue
completion_notes	TEXT	Evidence/notes
completed_at	TIMESTAMPTZ	Completion
evidence_id	UUID FK nullable	Completion evidence

Column	Type	Description
id	UUID PK	Effectiveness ID
capa_id	UUID FK UNIQUE	Parent CAPA
method	VARCHAR(120)	e.g. monitor_next_5_batches
criteria	TEXT	Success criteria
monitoring_start	DATE	Start
monitoring_end	DATE	End
status	VARCHAR(32)	planned/monitoring/effective/ineffective
result_summary	TEXT	Result
evidence	JSONB	Batch/metric evidence
reviewed_by	UUID FK	Reviewer
reviewed_at	TIMESTAMPTZ	Review time

Column	Type	Description
id	UUID PK	Review ID
batch_id	UUID FK UNIQUE	Batch
company_id	UUID FK	Tenant
reviewer_id	UUID FK	QA/authorized reviewer
status	VARCHAR(32)	pending/reviewed/approved/rejected
deviations_reviewed	BOOLEAN	All deviations reviewed?
investigations_complete	BOOLEAN	Investigations complete?
capa_review_complete	BOOLEAN	CAPA status reviewed
qc_results_status	VARCHAR(32)	pass/fail/pending
disposition	VARCHAR(32)	release/quarantine/reject
comments	TEXT	Decision rationale
decided_at	TIMESTAMPTZ	Decision timestamp
decided_by	UUID FK	Decision maker

Column	Type	Description
id	UUID PK	Complaint ID
reference	VARCHAR(32) UNIQUE	COM-2026-003
company_id	UUID FK	Tenant
product_name	VARCHAR(200)	Product
batch_id	UUID FK nullable	Related batch
deviation_id	UUID FK nullable	Related deviation
complaint_type	VARCHAR(48)	quality/adverse_event/other
description	TEXT	Complaint
status	VARCHAR(32)	open/investigating/closed
risk_level	VARCHAR(24)	Risk classification
recall_required	BOOLEAN	Recall flag
created_at	TIMESTAMPTZ	Timestamp

Column	Type	Description
id	UUID PK	Recall ID
reference	VARCHAR(32) UNIQUE	Recall reference
company_id	UUID FK	Tenant
complaint_id	UUID FK nullable	Triggering complaint
batch_id	UUID FK	Affected batch
reason	TEXT	Reason
status	VARCHAR(32)	assessment/initiated/closed
decision_notes	TEXT	Decision rationale

Column	Type	Description
id	UUID PK	Event
reference	VARCHAR(32) UNIQUE	AE reference
company_id	UUID FK	Tenant
complaint_id	UUID FK nullable	Related complaint
product_name	VARCHAR(200)	Product
batch_id	UUID FK nullable	Related batch
description	TEXT	Event description
seriousness	VARCHAR(32)	Seriousness
status	VARCHAR(32)	open/investigating/closed

Column	Type	Description
id	UUID PK	Supplier
company_id	UUID FK	Tenant
supplier_code	VARCHAR(64)	Code
name	VARCHAR(255)	Supplier name
status	VARCHAR(32)	approved/under_review/suspended
risk_rating	VARCHAR(24)	Low/Medium/High or configurable
last_audit_date	DATE	Audit date
created_at	TIMESTAMPTZ	Timestamp

Column	Type	Description
id	UUID PK	Event
supplier_id	UUID FK	Supplier
material_id	UUID FK nullable	Material
deviation_id	UUID FK nullable	Related deviation
batch_id	UUID FK nullable	Affected batch

Column	Type	Description
event_type	VARCHAR(48)	material_issue/audit/capa/complaint/other
description	TEXT	Quality event
status	VARCHAR(32)	open/investigating/closed
supplier_capa_reference	VARCHAR(64)	External/internal supplier CAPA

Column	Type	Description
id	UUID PK	AI interaction
company_id	UUID FK	Tenant
user_id	UUID FK	Actor
entity_type	VARCHAR(48)	deviation/investigation/capa/etc.
entity_id	UUID	Business entity
stage	VARCHAR(48)	extraction/triage/investigation/root_cause/capa/effectiveness
provider	VARCHAR(64)	LLM provider
model_name	VARCHAR(128)	Model
model_version	VARCHAR(128)	Version if available
prompt_version	VARCHAR(64)	Application prompt version
input_hash	VARCHAR(128)	Privacy-friendly dedup/traceability
recommendation	JSONB	Structured output
confidence	NUMERIC(5,4)	Model confidence if returned
human_decision	VARCHAR(32)	pending/accepted/edited/rejected
edited_fields	JSONB	Fields changed by human
created_at	TIMESTAMPTZ	Timestamp

Column	Type	Description
id	UUID PK	Link ID
ai_interaction_id	UUID FK	AI interaction
knowledge_document_id	UUID FK nullable	Source document
knowledge_chunk_id	UUID FK nullable	Retrieved chunk
source_type	VARCHAR(32)	SOP/batch_record/other
relevance_score	NUMERIC(6,5)	Retriever similarity/relevance
snippet	TEXT	Short evidence excerpt

Relationship	Cardinality	Delete behavior
Company → Sites	1:N	CASCADE site on company deletion only for non-GMP setup data; in production prefer deactivation
Company → Memberships	1:N	CASCADE membership
Batch → Batch Materials	1:N	Restrict/delete only in demo; never delete released batch history
Batch → Process Checks	1:N	Restrict after batch completion
Batch → Deviations	1:N	RESTRICT
Deviation → Investigation	1:0..1 for core workflow	RESTRICT
Investigation → Tasks	1:N	CASCADE only before finalization; otherwise retain history
Investigation → RCA	1:0..1	RESTRICT
Deviation → CAPA	1:N possible	RESTRICT
CAPA → Actions	1:N	RESTRICT
CAPA → Effectiveness	1:0..1	RESTRICT
Batch → Release Review	1:0..1	RESTRICT
Batch → Complaints	1:N	RESTRICT
Supplier → Materials	1:N	RESTRICT
Supplier → Quality Events	1:N	RESTRICT

Relationship	Cardinality	Delete behavior
Any entity → Audit Events	1:N	Audit events should never cascade-delete with business records

Table	Indexes to add	Why
batches	(company_id, status), (company_id, batch_number) UNIQUE	Dashboard + traceability
batch_materials	batch_id, material_id	Batch genealogy
manufacturing_steps	batch_id, step_number	Ordered process display
in_process_checks	(batch_id, checked_at), status	Trend and deviation trigger
deviations	(company_id, status), batch_id, owner_id, reference UNIQUE	Work queues
investigations	(company_id, status), deviation_id UNIQUE	Workflow navigation
investigation_tasks	investigation_id, owner_id, status	Task queues
root_cause_analyses	investigation_id UNIQUE	Fast investigation view
capas	(company_id, status), deviation_id, owner_id, reference UNIQUE	CAPA dashboard
capa_actions	capa_id, owner_id, status, due_date	Action tracking
effectiveness_checks	capa_id UNIQUE, status	Closure
complaints	(company_id, status), batch_id, deviation_id	Downstream traceability
suppliers	(company_id, status), supplier_code	Supplier lookup
supplier_quality_events	supplier_id, deviation_id, batch_id, status	Supplier investigation
ai_interactions	(company_id, entity_type, entity_id), created_at	AI history
ai_evidence_links	ai_interaction_id, knowledge_document_id, knowledge_chunk_id	Evidence inspection
audit_events	(company_id, entity_type, entity_id, created_at)	Audit history

Record	Example	Connection
Supplier	ChemCorp	Provides Acetic Anhydride
Raw Material	Acetic Anhydride / RM-2026-001	Linked to supplier
Batch	API-2026-041	Uses RM-2026-001
Process Step	Step 3 — Reaction	Batch process
In-Process Check	Temperature 84°C; spec 76–80°C	Creates deviation
Deviation	DEV-2026-018	Linked to batch/check
Investigation	INV-2026-012	Linked to deviation
Root Cause	Inadequate preventive-maintenance control	Linked to investigation
CAPA	CAPA-2026-009	Linked to deviation/investigation
Effectiveness	Monitor next 5 batches	Linked to CAPA
Batch Release	API-2026-041	Decision after quality review
Complaint	COM-2026-003	Linked to finished product/batch

Priority	Tables	Implementation depth
P0	deviations extensions, investigations, investigation_tasks, root_cause_analyses, capas, capa_actions, effectiveness_checks	Real CRUD + workflow transitions
P1	batches, manufacturing_steps, in_process_checks	Enough to create one API batch and trigger a deviation
P1	batch_release_reviews	One connected review screen
P1	suppliers, raw_materials, batch_materials	Traceability context
P1	complaints	One connected complaint record
P2	recalls, adverse_events	Minimal schema/UI if time permits
P2	ai_interactions, ai_evidence_links	Add if AI governance can be implemented without slowing core lifecycle